

CHAPTER 3:

PROGRAM CONTROL STATEMENTS

Overview

- Elements of Selection Structure to cover
  1. Definition of Selection Structure
  2. Type of Selection Structure
  3. Draw flowchart related to Selection Structure
  4. Write program with Selection Structure

Overview

- SELECTION statement - It tells the program to execute a certain section of code *only if* a particular test evaluates to true

- Variation of SELECTION statement
  - If statement
  - If - else statement
  - If - else if – else statement
  - Nested-if statement
  - Switch statement

If STATEMENT

'If' statement

syntax:

if (<conditional expression>)
<statement action>

Pseudocode:
If student's grade is greater than or equal to 60
Print "Passed"

Example:

```
public class IfDemo {
        public static void main(String[] args) {
            int grade;

            if (grade >= 60)
                    System.out.println("Passed");
        }
}
```

If Statement (examples)

- Examples

```
if (str_readdata.equals("1"))
{
        System.out.println(“Condition true”);
 }
```

- Examples

```
if (intdata == 5)
{
        System.out.println(“Condition true”);
 }
```

- Examples

```
if ( intdata >5        &&        intnumber <10 )
{
        System.out.println(“Condition true”);
 }
```

‘if’ structure flowchart

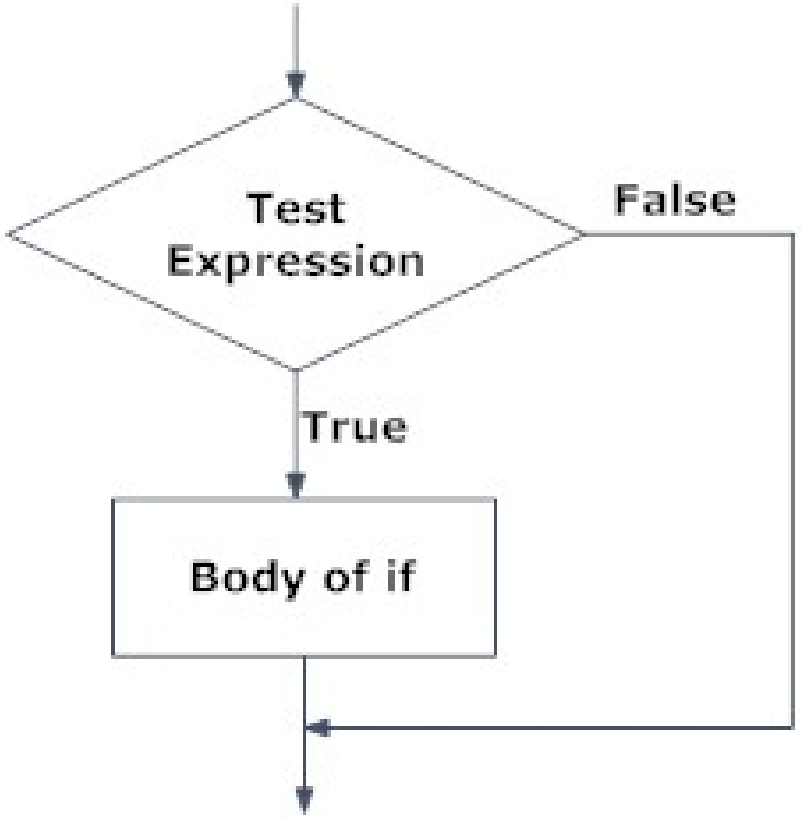


Fig: Operation of if statement

If - else STATEMENTS

'If…else' statement

syntax:

if (<conditional expression>)
<statement action>
else
<statement action>

Pseudocode:

If student's grade is greater than or equal to 60
Print "Passed"
else
Print "Failed"

Java Code:

```
public class IfDemo {
        public static void main(String[] args) {
            int grade;

            if (grade >= 60)
                System.out.println("Passed");
            else  if (    )
                System.out.println("Failed");
else if (      )
   sysyte   mm
else if (    )
            }
}
```

If - else Statement (examples)

- Examples

```
if (str_readdata.equals("1"))
{
        System.out.println(“Condition true”);
 }
else
{
         System.out.println(“Condition false”);
}
```

- Examples

```
if (intdata <=  34)
{
        System.out.println(“Condition true”);
 }
else
{
         System.out.println(“Condition false”);
}
```

If – else if -else Statement

- Examples

```
 if (str_readdata.equals("1"))
{
        System.out.println(“Number is 1”);
 }

else if  (intnumber == 1)
{
        System.out.println(“Number is 1”);
}

else if  (intnumber < 5 || intnumber >10 )
{
        System.out.println(“Number is  less than 5 bigger than 10”);
}

else
{
        System.out.println("Number not recognize");
}
```

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If – else if -else Statement

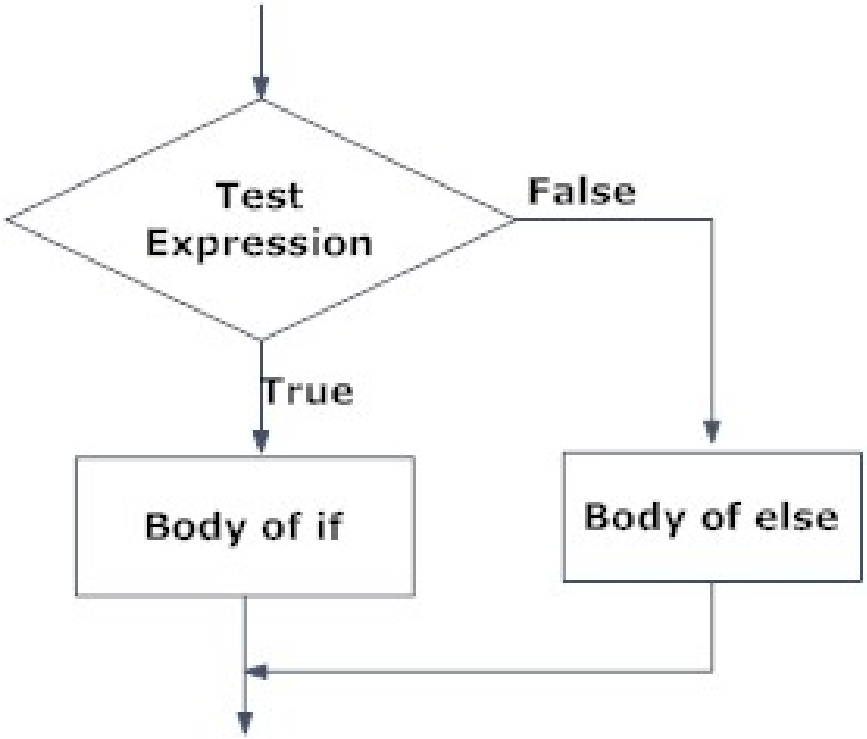


Fig: Operation of if...else statement

Nested if - Statement

- Examples

```
if (str_readdata.equals("1"))

{

        result = number1 * number2;
        if(result > 0)
        {
        System.out.println("You select multiply”);
        }
        else
        {
                        System.out.println("You select
divide”);
        }

}
```

SWITCH STATEMENTS (CONCEPTS)

- In the switch statement, the computer tests a variable consecutively against a list of integer or character constants.
- The switch statement is used to avoid long blocks of if..else code.
- After finding a match, the computer executes the statement or block of statements that are associated with the specified constant.
- the default statement is executed if the compiler does not find a match in the list of constants.
- each case must end with a ‘break’ statement

SWITCH STATEMENTS

'switch' statement

syntax:

switch (<non-long integral expression>) {
case $label_1$: <$statement_1$>
case $label_2$: <$statement_2$>
…
case $label_n$: <$statement_n$>
default: <statement>
} // end switch

SWITCH STATEMENTS (EXAMPLE)

Example:

```
class SwitchDemo {
        public static void main(String[] args) {
                int month = 14;
                switch (month) {
                        case 1: System.out.println("January"); break;
                        case 2: System.out.println("February"); break;
                        case 3: System.out.println("March"); break;
                        case 4: System.out.println("April"); break;
                        .
                        .
                        .
                        case 12: System.out.println("December"); break;
                        default: System.out.println("Invalid month.");break; } } }
```

SWITCH STATEMENTS (FLOW CHART)

